

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

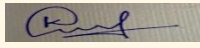
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	Kosicram K	Reg. No.:	192571079
Date:	01/10/2026	Duration:	60 minutes

Code of Conduct

I Kosicram K / 192571079 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
 - Justify why the selected methodology/framework is suitable for the proposed system.
 - Explain how it supports successful project development and maintenance.
- 6. Discuss Expected Outcomes and Limitations**
- Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
 - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework	Selects the most suitable SDLC/model/frame	Suitable methodology selected with	Methodology selected but	Outcomes, limitations, or future

Justification (10%)	work and provides strong technical justification.	adequate justification.	justification is weak.	enhancements are missing; report lacks organization and clarity.
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Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

Problem Statement:- AI-Based Resource and Operations Management System

1. Understand the Problem Statement

1.1 Explain the problem in your own words.

- The problem focuses on developing an AI-Based Resource and Operations Management System for ITBP to improve the management of operational activities, resources, and field information.
- The system will collect and process information from surveillance cameras, drones, sensors, and satellite imagery through a centralized platform.
- AI will be used to identify unusual activities and possible threats, provide alerts, and support faster operational decision-making.
- The system will also help with resource monitoring, predictive maintenance of equipment, patrol-route planning, and automated incident reporting.
- Overall, the solution aims to provide a secure, reliable, scalable, and integrated software system that supports ITBP personnel while keeping human decision-making as an important part of the process.

1.2 Identify the objectives of the proposed system.

The major objectives of the proposed system are:

1. To develop an AI-based platform for analyzing operational data from multiple sources.
2. To improve situational awareness by identifying potential threats and unusual activities.
3. To optimize the utilization of available personnel, equipment, and other operational resources.
4. To support predictive maintenance of equipment and infrastructure.
5. To assist in optimizing patrol routes based on available operational information.
6. To automate the recording and reporting of operational incidents.

1.3 State the Scope of the Problem

The scope of the proposed system includes the development of a centralized software platform consisting of data collection, AI-based analysis, resource management, alert management, route optimization, incident reporting, and administrative modules.

The system will process data received from sources such as surveillance cameras, drones, sensors, and satellite imagery. AI-based modules will analyze the available data to identify anomalies and potential threats and provide relevant information to authorized users.

The system will also support resource monitoring, predictive maintenance, patrol-route planning, and automated generation of incident reports.

The proposed system primarily focuses on **software-based analysis, coordination, monitoring, and decision support**. It is not intended to replace human decision-making; instead, it provides authorized personnel with relevant information and analytical support for operational decisions.

2. Identify Key Stakeholders and Challenges

2.1 & 2.2 Identify all the stakeholders involved in the system & explain the roles and responsibilities of each stakeholder.

Stakeholder	Role and Responsibility
ITBP Personnel	Use system-generated information for monitoring and operational activities.
Command and Control Officers	Monitor operational information, review alerts, and support decision-making.
Operations Managers	Manage resources, personnel, equipment, and operational activities.
Technical/System Administrators	Maintain the software platform, user accounts, system configurations, and integrations.
Maintenance Personnel	Monitor equipment and infrastructure conditions and respond to maintenance requirements.
System Developers	Design, develop, test, deploy, and maintain the software system.

2.3 Discuss the major challenges and issues faced by the stakeholders.

Large Volume of Data

- The system may receive information from multiple sources, including cameras, drones, sensors, and satellite imagery. Processing and organizing such heterogeneous data efficiently is a major challenge.

Real-Time or Timely Analysis

- Operational decisions may depend on receiving relevant information quickly. Delays in data processing or alert generation can reduce the usefulness of the system.

Difficult Operating Environment

- The system must function in remote areas and under harsh environmental conditions. Therefore, reliability and availability are important requirements.

Data Integration

- Different information sources may use different formats and interfaces. Integrating these sources into a single software platform requires well-designed APIs and data-processing mechanisms

3. Analyze the Current Approach

3.1 Describe the existing system or current process

The problem statement indicates that ITBP operates across extensive and challenging geographical regions and requires information from multiple operational sources. In a conventional or fragmented approach, information from different sources may need to be monitored, analyzed, and coordinated through separate systems or processes.

Such an approach can make it difficult to obtain a unified operational view. Data from surveillance systems, drones, sensors, satellite imagery, equipment status, and operational reports may not always be available through a single integrated platform.

The existing approach can therefore be considered as a combination of monitoring systems, operational procedures, human analysis, and reporting mechanisms rather than a unified AI-assisted operational management platform.

3.2 Identify its strengths and limitations

- Human personnel can apply operational knowledge and experience.
- Existing systems and procedures can continue to support day-to-day operations.
- Different sources can provide specialized information for different operational requirements.
- Human judgment remains an important component of decision-making.

3.3 Identified Gaps

The major gaps that the proposed system should address are:

1. Lack of centralized analysis of heterogeneous operational data.
2. Limited automated detection of anomalies and potential threats.
3. Limited predictive analysis for equipment and infrastructure maintenance.
4. Need for improved resource utilization.

4. Propose a suitable Solution Using Software Engineering Principles

4.1 Design a software-based solution for the given problem

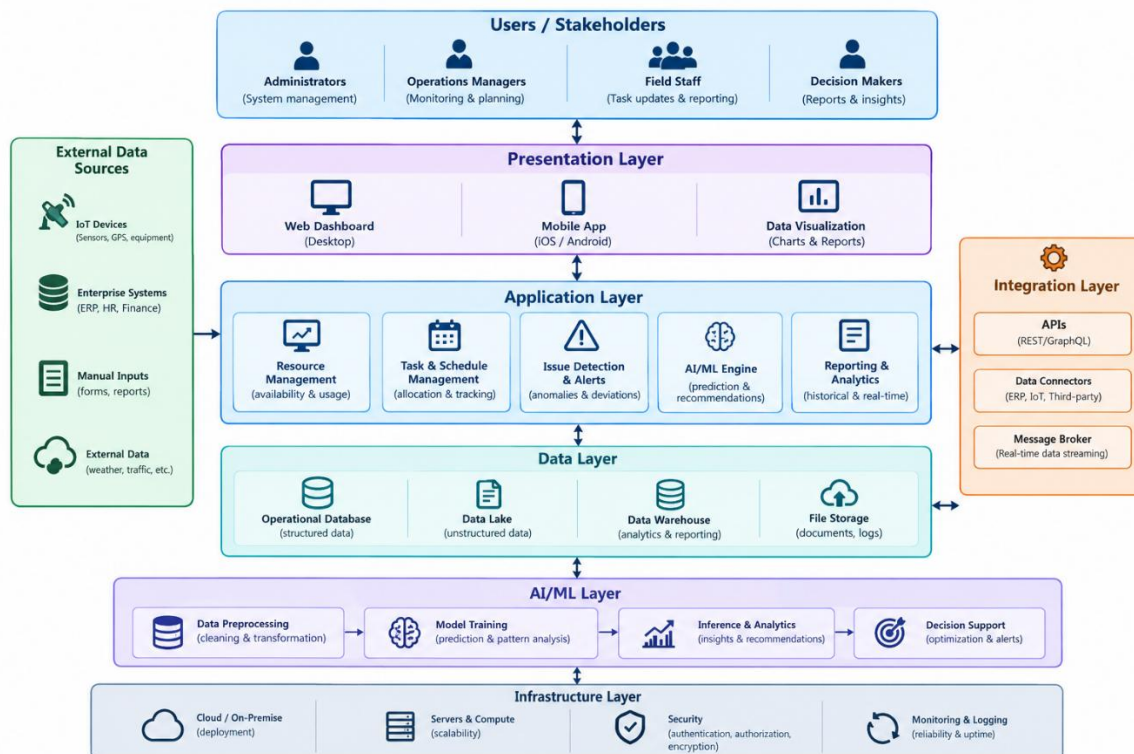
The proposed **AI-Based Resource and Operations Management System** will be a centralized, modular software platform that collects operational data from multiple sources, processes the data, applies AI-based analytical techniques, and presents relevant information through dashboards and alerts.

The system will use a layered architecture consisting of a user interface layer, application/service layer, AI and analytics layer, data-management layer, and integration layer.

Data Sources → Data Processing → AI Analysis → Decision Support → Alerts/Dashboard → Authorized Personnel

4.2 Describe the major functional modules of the proposed system.

- 1. User and Access Management Module**-Manages user registration, authentication, roles, and access permissions.
- 2. Data Integration Module**-Collects and integrates resource, workforce, equipment, task, and operational data from different sources.
- 3. AI-Based Threat and Anomaly Detection Module**-Analyzes operational data to detect inefficiencies, predict resource requirements, and generate recommendations.
- 4. Resource Management Module**-Monitors resource availability and usage while supporting efficient allocation and utilization.



4.3 Identify the functional and non-functional requirements.

The system shall:

1. Allow authorized users to securely access the platform.
2. Collect and process information from supported data sources.
3. Analyze operational data using AI-based techniques.
4. Identify anomalies and potential threats.
5. Generate alerts based on detected events.

Non-Functional Requirements

- **Security:** The system should protect operational data using authentication, authorization, encryption, and secure communication.
- **Reliability:** The system should provide dependable services and handle failures without unnecessary data loss.
- **Availability:** Important system functions should remain accessible whenever required.
- **Performance:** Data processing, dashboard loading, and alert generation should occur within acceptable response times.
- **Interoperability:** The system should support integration with existing command-and-control systems and external data sources.

4.3 Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution

- **Modularity:** The system is divided into independent modules such as user management, AI analysis, resource management, route optimization, and incident management.
- **Scalability:** A scalable architecture and service-based design can allow additional users, data sources, and processing requirements to be incorporated.
- **Maintainability:** Clear separation of modules, documentation, coding standards, automated testing, and version control will improve maintainability.
- **Reliability:** Error handling, logging, monitoring, backup mechanisms, and fault-tolerant components can improve system reliability.
- **Usability:** Role-specific dashboards and simple interfaces will reduce unnecessary complexity for end users.
- **Security:** Authentication, role-based authorization, encryption, secure APIs, audit logs, and controlled access will protect sensitive information.
- **Reusability:** Common services such as authentication, notification, logging, and reporting can be designed as reusable components.

5. Justification of the Chosen

Methodology/Framework

Selected Methodology: Agile Scrum

Agile Scrum is selected as the software development methodology for the proposed system.

The system contains multiple modules and involves different categories of stakeholders. Requirements may evolve as users evaluate the system and provide feedback. Therefore, developing the entire system in a single development cycle may create difficulties in identifying and correcting requirements early.

Agile Scrum supports **incremental and iterative development**, allowing the system to be divided into manageable modules and developed through multiple sprints.

Proposed Sprint Structure

Sprint Major Activities

Sprint 1 Requirement analysis, user management and system architecture

Sprint 2 Data integration and data-processing module

Sprint 3 AI-based anomaly and threat analysis

Sprint 4 Resource management and predictive maintenance

Supporting DevOps Practices

Agile development can be supported by DevOps practices such as:

- Version control
- Continuous Integration
- Automated testing
- Continuous Delivery/Deployment
- Application monitoring
- Automated build and deployment processes

This combination can help maintain software quality throughout development and maintenance.

6. Discuss Expected Outcomes and Limitations

6.1 Describe the expected benefits of the proposed solution.

The proposed system is expected to:

1. Improve the centralized management of operational information.
2. Enhance situational awareness through AI-assisted analysis.
3. Support faster identification of anomalies and potential threats.
4. Improve utilization of available resources.

6.2 Explain how the solution addresses the identified challenges.

The system addresses fragmented information by providing a centralized platform for data integration and analysis.

Manual analysis is reduced by using AI-based processing for anomaly detection and operational analytics. Resource-utilization challenges are addressed through a dedicated resource-management module and data-driven decision support.

Maintenance challenges are addressed through predictive maintenance capabilities.

Operational reporting is improved through automated incident recording and report generation.

6.3 Discuss potential risks, implementation challenges, and limitations.

Risk	Mitigation
Poor-quality input data	Data validation and preprocessing
Incorrect AI predictions	Human verification and continuous model evaluation
Cybersecurity threats	Strong authentication, authorization, encryption and monitoring
System integration difficulties	Standard APIs and phased integration

6.3 Limitations

- AI performance depends on the quality and availability of training and operational data.
- AI-based predictions may not always be completely accurate.
- Integration with existing systems may require significant technical effort.
- Processing large volumes of multimedia and sensor data can require substantial computing resources.

6.4 Suggest possible Future Enhancements

Future versions of the system could include:

1. Advanced multimodal AI capable of jointly analyzing video, image, sensor, and other operational data.
2. Improved predictive analytics for resource and maintenance planning.
3. Advanced geospatial visualization and mapping.
4. Integration with additional IoT and sensor platforms.

Conclusion

The proposed AI-based system integrates operational data, resource management, threat detection, maintenance, and reporting into a single secure platform, improving situational awareness and decision-making through Software Engineering principles, while maintaining human oversight in critical operations.